Python Chapter 6 Dictionaries Activities

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Time required: 60 minutes

How to Think Like a Computer Scientist (Interactive Edition)

Go through the following tutorials.

Dictionaries

Online Tutorials

Go through the following tutorials.

- Python Dictionaries
    - Access Items
    - Change Items
    - Add Items
    - Remove Items
    - Loop Dictionaries
    - Copy Dictionaries
    - Nested Dictionaries

- Dictionary Methods

Tutorial 1: Cats and Dogs

A simple dictionary example with cats and dogs.

```python
"""
    Name: cats_and_dogs.py
    Author:
    Created:
    Purpose: Create and use a dictionary
"""
# Define the key : value pairs of the dictionary
dictionary = {
    "dog": "has a tail and goes woof!",
    "cat": "says meow",
    "mouse": "is chased by cats"
}

# Prompt the user to enter a dictionary key
print("This dictionary contains values for dog, cat, or mouse.")
word = input("Enter a word (key): ")

# Use the key entered by the user to access the value
# .get("key", "default value") default value is used if key doesn't exist)
print(f"Key: {word} Value: {dictionary.get(word, 'Value does not exist')}")
print(f"A {word} {dictionary.get(word, 'Value does not exist')}")
```

Example run:

```
This dictionary contains values for dog, cat, or mouse.
Enter a word (key): dog
The value associated with dog: has a tail and goes woof!
A dog has a tail and goes woof!
```

```
This dictionary contains values for dog, cat, or mouse.
Enter a word (key): key
Key: key Value: Value does not exist
A key Value does not exist
```

Tutorial 2: Add Items to a Dictionary

```python
"""
    Name: product_price_dictionary_2.py
    Author: William A Loring
    Created: 02/23/2022
    Purpose: Product name and price dictionary
"""

# Create empty dictionary
product_dict = {}

while True:
    # Get item from user
    product_name = input("Enter product name: ")
    product_price = float(input("Enter product price: "))

    # Insert item into dictionary
    product_dict[product_name] = product_price

    # Print the dictionary directly
    print(product_dict)

    # Print the dictionary in a nicer format
    # for each items key and value
    # loop through the dictionary
    for key, value in product_dict.items():
        print(f"{key}: {value}")

    choice = input("Enter another item? (y) (Enter to exit) ")
    if choice == "":
        break
```

Example run:

```
Enter product name: Ice Cream
Enter product price: 7.99
{'Ice Cream': 7.99}
Ice Cream: 7.99
Enter another item? (y) (Enter to exit) y
Enter product name: Sprinkles
Enter product price: 1.99
{'Ice Cream': 7.99, 'Sprinkles': 1.99}
Ice Cream: 7.99
Sprinkles: 1.99
```

Tutorial 3: Pickle a Dictionary

Update the previous tutorial to pickle and unpickle a dictionary.

```python
"""
    Name: product_dictionary_pickle.py
    Author: William A Loring
    Created: 10/08/2023
    Purpose: Pickle product and price dictionary
"""
import pickle
FILE_NAME = "product_dictionary.pkl"
# Create empty dictionary object
product_dict = {}

"""Unpickle the dictionary from file with pickle.load
    'with open' opens the file for access
    'r' read file
    'b' binary file type
"""
# Use try catch for exception if the file doesn't exist
try:
    with open(FILE_NAME, "rb") as file_handle:
        product_dict = pickle.load(file_handle)
    # When the program exits the 'with' block,
    # the file is closed: the file handle resource is released
    print("Load pickle dictionary")
    # Print the dictionary
    for product, price in product_dict.items():
        print(f"{product}: {price}")
except:
    pass
```

```python
while True:
    # Get item from user
    product_name = input("Enter product name: ")
    product_price = float(input("Enter product price: "))

    # Insert item into dictionary using 'product_name' as the key
    product_dict[product_name] = product_price

    """Pickle the dictionary to a file with pickle.dump
        'with open' opens the file for access
        'w' write file
        'b' binary file type
    """
    with open(FILE_NAME, "wb") as file_handle:
        # Write list to file with binary protocol
        pickle.dump(product_dict, file_handle)
    # When the program exits the 'with' block,
    # the file is closed: the file handle resource is released

    print("Dump pickle dictionary")
    # Print the dictionary
    for product, price in product_dict.items():
        print(f"{product}: {price}")

    choice = input("Enter another item? (y) (Enter to exit) ")
    if choice == "":
        break
```

Example run:

```
Load pickle dictionary
Carrots: 2.99
beans: 4.5
corn: 3.4
Enter product name: radishes
Enter product price: 2.34
Dump pickle dictionary
Carrots: 2.99
beans: 4.5
corn: 3.4
radishes: 2.34
Enter another item? (y) (Enter to exit)
```

Assignment 1: Che Carlton's Menu

This example uses various dictionary methods in a point-of-sale program.

Use a menu to add pickling to the program to load and dump the CafeMenu object.

```python
"""
    Name: cafe_menu_dictionary.py
    Author:
    Created:
    Purpose:
"""


class CafeMenu:
    def __init__(self):
        # Define cafe menu dictionary
        self.cafe_menu = {
            "Hamburger": 5.25,
            "Apple Pie": 6.00,
            "Bacon and Eggs": 5.95,
            "Biscuits and Gravy": 7.25
        }
```

```python
        # Print out the menu
        self.print_cafe_menu()

        # Print a specific menu price by key
        print(f"\nDisplay a specific menu item's price.")
        # Get the price of the Hamburger from the dictionary
        hamburger = self.cafe_menu.get('Hamburger')
        print(
            f"Che Carlton's speciality is Hamburgers: {hamburger}")
```

```python
        # Delete a key-value pair
        print(f"Delete Hamburger from the menu.")
        del self.cafe_menu["Hamburger"]

        # Print the number of menu items
        print(
            f"\nThere are {len(self.cafe_menu)} items in the menu")

        # Print out the menu
        self.print_cafe_menu()

        # Adding a key-value pair
        print(f"\nAdd Hamburger Deluxe to the menu.")
        self.cafe_menu["Hamburger Deluxe"] = 6.25

        print(f"\nIs Hamburger Deluxe in the menu?.")
        if "Hamburger Deluxe" in self.cafe_menu:
            print(f"Hamburger Deluxe: {self.cafe_menu.get('Hamburger')}")

        self.print_cafe_menu()

    def print_cafe_menu(self):
        """ Print out the menu """
        print("\n----------------------------")
        print(f"Che Cartlon's Cafe Menu")
        print("----------------------------")

        # Go through the dictionary, print one item at a time
        for menu_item, price in self.cafe_menu.items():
            print(f"{menu_item}: ${price}")


# Create program object, start program
cafe_menu = CafeMenu()
```

Example run:

```
----------------------------
Che Cartlon's Cafe Menu
----------------------------
Hamburger: $5.25
Apple Pie: $6.0
Bacon and Eggs: $5.95
Biscuits and Gravy: $7.25

Display a specific menu item's price.
Che Carlton's speciality is Hamburgers: 5.25

There are 4 items in the menu.
Delete Hamburger from the menu.

There are 3 items in the menu

----------------------------
Che Cartlon's Cafe Menu
----------------------------
Apple Pie: $6.0
Bacon and Eggs: $5.95
Biscuits and Gravy: $7.25

Add Hamburger Deluxe to the menu.

Is Hamburger Deluxe in the menu?.
Hamburger Deluxe: None

----------------------------
Che Cartlon's Cafe Menu
----------------------------
Apple Pie: $6.0
Bacon and Eggs: $5.95
Biscuits and Gravy: $7.25
Hamburger Deluxe: $6.25
```

Assignment Submission

1. Attach the program files.
2. Attach screenshots showing the successful operation of the program.
3. Submit in Blackboard.